

Task:

Choose any company or website such as Amazon, Flipkart, Netflix, Instagram, Facebook, YouTube, Swiggy, Zomato, etc.

Prepare a report containing:

- 1. Company Name - Instagram**
- 2. What Services Does the Company Provide?**

Instagram is a social networking platform that allows users to:

- Share photos and videos
- Create Stories and Reels
- Send direct messages (DMs)
- Follow friends, celebrities, and businesses
- Live stream videos
- Advertise products and services
- Create and manage business accounts

- 3. What User Data Might the Company Collect?**

Personal Information

- Name
- Username
- Date of Birth
- Gender

Contact Information

- Email Address
- Phone Number

Location Information

- Address (if provided)
- GPS Location
- IP Address

Payment Information

- Debit/Credit Card Details
- Billing Address
- Transaction History

Other Information

- Photos and Videos uploaded
- Messages and Chats
- Device Information
- Search History
- Followers and Following List
- Browsing Activity
- Login Records

4. What Cyber Attacks Could Target Instagram?

1. Phishing Attack

Attackers create fake Instagram login pages to steal user credentials.

2. Data Breach

Hackers gain unauthorized access to databases containing user information.

3. Ransomware Attack

Malware encrypts company data and demands payment for recovery.

4. Account Takeover

Attackers obtain usernames and passwords and gain control of user accounts.

5. DDoS (Distributed Denial of Service) Attack

A large amount of fake traffic is sent to Instagram's servers, making the service unavailable.

5. What Would Happen if Instagram Were Hacked?

If Instagram were successfully hacked:

- Millions of user accounts could be compromised.
- Personal information could be leaked.
- Private messages and photos could become public.
- Users may lose access to their accounts.
- Financial information could be stolen.
- The company's reputation would be damaged.
- Users could lose trust in the platform.
- The company could face legal penalties and financial losses.

6. Suggest at Least 5 Security Measures to Protect the Company

1. Multi-Factor Authentication (MFA)

Require users and employees to verify identity using an additional authentication method.

2. Strong Password Policy

Enforce complex passwords and regular password updates.

3. Data Encryption

Encrypt user data during storage and transmission.

4. Regular Security Audits

Conduct vulnerability assessments and penetration testing.

5. Employee Cybersecurity Training

Train employees to recognize phishing emails and security threats.

6. Network Monitoring

Monitor network activity continuously for suspicious behavior.

7. Regular Backups

Maintain secure backups to recover data after cyber incidents.

7. If You Were the Cyber Security Manager, What Would Be Your First 3 Actions?

Action 1: Strengthen Access Control

Implement Multi-Factor Authentication (MFA) for all employees and administrators and review user access permissions.

Action 2: Conduct Security Assessment

Perform vulnerability scanning and penetration testing to identify weaknesses in the system.

Action 3: Establish Continuous Monitoring

Deploy security monitoring tools, intrusion detection systems, and incident response procedures to quickly detect and respond to attacks.

8. Conclusion

Instagram is one of the world's largest social media platforms and stores a vast amount of user information. It faces threats such as phishing, data breaches, ransomware, account takeovers, and DDoS attacks. Implementing strong security controls, employee awareness programs, and continuous monitoring can significantly reduce cyber risks and protect both the company and its users.